SVM/SVC Summary

idea: Find class-separating hyperplanes $\{\vec{x} \in \mathbb{R}^p : \underbrace{\beta_0 + \vec{x}^T\vec{\beta}}_{f(\vec{x})} = 0\}$ (MMC)

$> \Rightarrow y_i = +1$

$< \Rightarrow y_i = -1$

If not separable $\Rightarrow$ find one with distances to misclassified pts. "as small as possible" (SVC)

$\Downarrow$

$$\vec{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

SVC "Budget" margin optimization

$$\boxed{\max_{\beta_0, \vec{\beta}, \epsilon_1, \dots, \epsilon_n} M \quad \text{s.t.:} \begin{cases} \sum_{j=1}^{p} \beta_j^2 = 1 \\ y_i(\beta_0 + \vec{x}_i^T\vec{\beta}) \geq M(1-\epsilon_i) \\ \epsilon_i \geq 0, \; \sum_{i=1}^{n} \epsilon_i \leq b \qquad \forall i = 1, \dots, n \end{cases}}$$

$\Longrightarrow$

Solution $f^*(\vec{x}) = \beta_0^* + \vec{x}^T\vec{\beta}^*$ is of the form:

$$\boxed{f^*(\vec{x}) = \beta_0^* + \sum_{i=1}^{n} \alpha_i y_i \langle x, x_i \rangle}$$

for some $\alpha_1, \dots, \alpha_n$

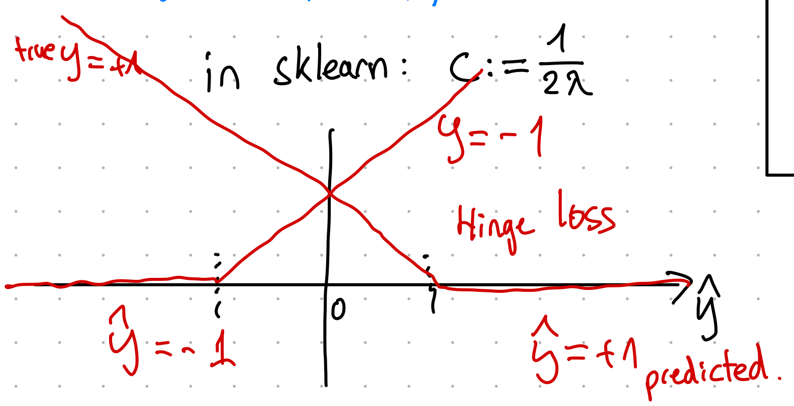
$\Downarrow$ $\quad\beta_0, \vec{\beta}\ \Updownarrow \quad\nearrow$

SVC penalized loss optimization

$$\boxed{\min_{\beta_0, \vec{\beta}} \sum_{i=1}^{n} [1 - y_i(\underbrace{\beta_0 + \vec{x}_i^T\vec{\beta}}_{\hat{y}})]_+ + \lambda \|\vec{\beta}\|_2^2}$$

"HingeLoss" $(y_i, \beta_0 + \vec{x}_i^T\vec{\beta})$ $\qquad$ L_2 penalty

in sklearn: $C := \frac{1}{2\lambda}$



Kernel trick: swap $\langle \cdot, \cdot \rangle$ for a non-linear kernel $K(\cdot, \cdot)$

$\Updownarrow$

Equivalent to working in a higher-dimensional feature space, as $K(\cdot, \cdot) = \langle \phi(\cdot), \phi(\cdot) \rangle$ for some transformation

$\phi : \mathbb{R}^p \to \mathbb{R}^q$

of the features

3. Theoretical Exercise: Kernels

Let $x_1, x_2 \in \mathbb{R}^3$ be observations in three dimensions (vectors). Consider the kernel function $k : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$ defined as $k(x_1, x_2) = (1 + x_1^T x_2)^2$.

With this exercise, we want to show that any valid kernel *implicitly* represents a dot product between two *transformed* vectors. In other words, for any kernel k, one can find a transformation of the variables $\phi(x)$ (i.e., a feature transformation) such that $k(x_1, x_2) = \phi(x_1)^T \phi(x_2)$. $= \langle \phi(x_1); \phi(x_2) \rangle$

This result helps a lot because one can represent a very high-dimensional feature transformation without having to compute $\phi(x)$ explicitly (more about this in the solutions).

1. Find a feature transformation $\phi(x)$, for $x \in \mathbb{R}^3$, such that $k(x_1, x_2) = \phi(x_1)^T \phi(x_2)$.
2. We know that a feature transformation is a function $\phi : \mathbb{R}^3 \to \mathcal{X}$ that maps each observation $x \in \mathbb{R}^3$ to some point $\phi(x)$ in another space $\mathcal{X}$. What is the dimension of such space $\mathcal{X}$?

① Let $x, v \in \mathbb{R}^3$ be two vectors in the feature space

$$k(x, v) = (1 + x^T v)^2 = (1 + x_1 v_1 + x_2 v_2 + x_3 v_3)^2$$

$$= (1 + x_1 v_1 + x_2 v_2 + x_3 v_3)(1 + x_1 v_1 + x_2 v_2 + x_3 v_3)$$

$$\begin{aligned} = 1 &+ 2x_1 v_1 + 2x_2 v_2 + 2x_3 v_3 \\ &+ (x_1 v_1)^2 + (x_2 v_2)^2 + (x_3 v_3)^2 \\ &+ 2(x_1 x_2)(v_1 v_2) + 2(x_1 x_3)(v_1 v_3) + 2(x_2 x_3)(v_2 v_3) \end{aligned}$$

$\phi(x)^T \phi(v)$

$\phi(x)^T$

$\shortparallel$

$$\begin{aligned} = &[1, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_3, x_1^2, x_2^2, x_3^2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_3, \sqrt{2}x_2 x_3] \\ &\cdot [1, \sqrt{2}v_1, \sqrt{2}v_2, \sqrt{2}v_3, v_1^2, v_2^2, v_3^2, \sqrt{2}v_1 v_2, \sqrt{2}v_1 v_3, \sqrt{2}v_2 v_3]^T \end{aligned}$$

$= \phi(v)$

$$= \phi(x)^T \phi(v).$$

②

$$\phi : \mathbb{R}^3 \longrightarrow \mathcal{X} \subset \mathbb{R}^{10}$$

$\Rightarrow$ dim. of $\mathcal{X}$ is 10

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \longmapsto \begin{pmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ \sqrt{2}x_3 \\ x_1^2 \\ x_2^2 \\ x_3^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2}x_1x_3 \\ \sqrt{2}x_2x_3 \end{pmatrix}$$